# PRISMSPACE – INTELLIGENT MULTI-AGENT PLATFORM

**Dept. of Computer Science & Engineering**
College of Engineering, Karunagappally

## Project Group 01

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|---|---|
| **NOBIN SIJO** | KNP23CS082 |
| **PRANAV P** | KNP23CS085 |
| **PRASANTH P** | KNP23CS086 |
| **SANKAR NARAYAN S** | KNP23CS094 |

*Project Coordinator & Guide*

**Dr. Leena Silvoster**
Associate Professor
College of Engineering, Karunagappally

# 01. Introduction: PrismSpace Web Platform

- A multi-agent system breaks complex goals into smaller tasks, assigns them to suitable agents, runs independent tasks in parallel, and combines their results.
- **PrismSpace** is a web-based multi-agent intelligence platform designed to coordinate AI agents and manage complex workflows.
- It converts natural-language goals into **Directed Acyclic Graphs (DAGs)** to manage task dependencies and enable parallel execution.
- It provides real-time workflow visualization, local session storage, and human approval controls for sensitive actions.

# Problem Statement

Challenges in current AI-agent and workflow systems

# 02. Problem Statement: Core Challenges in AI Agent Systems

- **Problem:** Coordinating multiple AI agents and complex tasks requires efficient task orchestration, seamless tool integration, clear workflow visibility, and secure execution.
- **Input:** Developer task prompts, workspace and session state, uploaded files, and MCP and human approval configurations.
- **Method:** PrismSpace applies intent classification, supervised model routing, DAG decomposition with topological scheduling, MCP-based tool execution, ORPO preference alignment, and human approval for sensitive actions.
- **Output:** Task results, tool outputs, validation results, and workflow status summaries.

# 03. Objectives

1. **Intent Classification Module:** Identifies user intent and classifies incoming prompts.
2. **DAG Generation Module:** Decomposes user goals into dependency-aware subtasks and schedules their execution.
3. **Agent Routing Module:** Assigns tasks to suitable agents based on task requirements.
4. **MCP Tool Integration Module:** Connects approved external tools and manages tool execution.
5. **Validation and Human Approval Module:** Validates outputs and requests human approval for sensitive actions.
6. **Workflow Visualization and Session Management Module:** Displays real-time task progress and stores session data locally.

# 04. Literature Review

Research background, key findings, and identified gap

# Multi-Objective Reward Generalization in Deep Reinforcement Learning

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **F. Cornalba, C. Disselkamp, D. Scassola, and C. Helf** *Neural Computing and Applications* **Year:** 2024 **Publisher:** Springer Open Access | Investigates **Multi-Objective Deep Reinforcement Learning (MODRL)** for optimizing competing operational objectives (e.g., return vs. risk). Introduces reward generalization techniques that train pareto-optimal policies capable of dynamically adapting to shifting stakeholder preferences without retraining from scratch. | **Advantages:**<br>▶ Provides structured Pareto-optimal policy exploration across conflicting objectives.<br>▶ Generalizes across parameterized reward weights without full policy retraining.<br>▶ Directly informs multi-objective routing (latency vs. cost vs. quality) in PrismSpace.<br>**Disadvantages:**<br>▶ High initial training sample complexity across multi-dimensional reward spaces.<br>▶ Evaluated primarily on continuous financial time-series data.<br>▶ Sensitive to hyperparameter tuning in policy entropy regularization. |

# Average Reward Adjusted Discounted Reinforcement Learning

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **M. Schneckenreither and G. Moser**<br>*Neural Computing and Applications*<br>**Year:** 2025<br>**Publisher:** Springer<br>Open Access | Proposes **Average-Reward Adjusted Discounted RL (Reinforcement Learning) (ARAD-RL)**, bridging the gap between discounted and average reward criteria in infinite-horizon **MDPs** (Markov Decision Processes). Addresses systematic value estimation bias in standard discounted Q-learning by dynamically subtracting the estimated gain rate during Bellman updates. | **Advantages:**<br>▶ Eliminates infinite-horizon return explosion while preserving policy convergence.<br>▶ Improves long-term steady-state throughput in continuous scheduling tasks.<br>▶ Theoretically sound formulation for continuous agent lifecycle optimization.<br>**Disadvantages:**<br>▶ Requires accurate empirical estimation of average reward baseline.<br>▶ Slow convergence in highly non-stationary transition environments.<br>▶ Adds variance to early-stage **TD** (Temporal Difference) error gradient estimates. |

# Deep Reinforcement Learning for Multi-Agent Coordination

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **K. Aina and S. Ha**<br>*Artificial Life and Robotics*<br>**Year:** 2026<br>**Publisher:** Springer Open Access | Develops decentralized **MARL** (Multi-Agent Reinforcement Learning) algorithms for multi-agent coordination in constrained environments. Employs **spatial-temporal observation encoding** and localized interaction graphs to coordinate joint movements and resolve deadlocks without requiring centralized global state knowledge. | **Advantages:**<br>▶ Resolves shared-resource contention and deadlocks via decentralized policies.<br>▶ Spatial-temporal graph attention scales linearly with agent density.<br>▶ Applicable to concurrent tool-use contention across worker agents.<br>**Disadvantages:**<br>▶ Tested in physical robotics grids rather than software **API** (Application Programming Interface) environments.<br>▶ Vulnerable to communication packet dropouts and sensor latency.<br>▶ Does not account for semantic language-based coordination. |

# Open-Ended Coordination Using Modular Open Policies

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **D. Rother, J. Pajarinen, J. Peters, and T. H. Weisswange** *Autonomous Agents and Multi-Agent Systems* **Year:** 2025 **Publisher:** Springer Open Access | Presents a framework for **open-ended ad hoc multi-agent coordination**. Utilizes **Modular Open Policies (MOP)** where dynamic agent teams dynamically infer unseen teammate skill representations and synthesize blended coordination strategies without requiring pre-coordinated joint training. | **Advantages:**<br>▶ Enables zero-shot ad hoc coordination with unfamiliar teammate agent policies.<br>▶ Modular policy decomposition allows plug-and-play agent role assignment.<br>▶ Directly adopted in PrismSpace for dynamic worker swarm scaling.<br>**Disadvantages:**<br>▶ Requires comprehensive pre-trained skill policy libraries for base roles.<br>▶ Evaluated primarily on benchmark games (Overcooked, **LBF** (Level-Based Foraging)).<br>▶ Action blending may introduce transient sub-optimal execution steps. |

# Improving Scalability of **MADRL** (Multi-Agent Deep Reinforcement Learning) with Suboptimal Human Knowledge

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **D. Liu, F. Ren, J. Yan, G. Su, W. Gu, and S. Kato**<br>*Autonomous Agents and Multi-Agent Systems*<br>**Year:** 2026<br>**Publisher:** Springer Open Access | Addresses sample inefficiency in large-scale **Multi-Agent DRL** (Deep Reinforcement Learning) by integrating heuristic human guidance into exploration. Proposes an adaptive filtering mechanism that extracts actionable macro-actions from noisy or sub-optimal human demonstration traces, accelerating multi-agent policy learning. | **Advantages:**<br>▶ Accelerates policy exploration by up to 4x using noisy demonstration priors.<br>▶ Robust to suboptimal or conflicting human heuristic rules.<br>▶ Informs PrismSpace's bootstrap initialization from logged developer sessions.<br>**Disadvantages:**<br>▶ Quality of learned policies depends on initial heuristic coverage.<br>▶ Requires complex threshold tuning to transition away from demonstrations.<br>▶ High memory footprint for maintaining dual replay buffers. |

# $\varepsilon$-Retraining Reinforcement Learning Algorithms

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **L. Marzari, C. Liu, P. L. Donti, and E. Marchesini** *Autonomous Agents and Multi-Agent Systems* **Year:** 2026 **Publisher:** Springer Open Access | Investigates bounded policy fine-tuning through $\varepsilon$-**retraining**, preventing catastrophic forgetting when deep **RL** agents adapt to non-stationary environments. Establishes formal Lipschitz bounds on policy drift, ensuring safe continual learning in deployed agent pipelines. | **Advantages:**<br>▶ Provides mathematical guarantees against catastrophic policy divergence.<br>▶ Enables continuous telemetry-driven model fine-tuning in production.<br>▶ Crucial for stability in PrismSpace's ongoing user interaction learning.<br>**Disadvantages:**<br>▶ Constrained optimization increases computation per gradient step.<br>▶ Conservative update bounds may slow adaptation to drastic distribution shifts.<br>▶ Theoretical guarantees rely on local smoothness assumptions. |

# Deep Multiagent Reinforcement Learning: Challenges and Directions

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **A. Wong, T. Bäck, A. V. Kononova, and A. Plaat**<br>*Artificial Intelligence Review*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Surveys **Deep Multiagent Reinforcement Learning (MARL)**, formalizing foundational challenges including non-stationarity, multi-agent credit assignment, and combinatorial action explosion. Evaluates **Centralized Training with Decentralized Execution (CTDE)** and communication architectures for modular orchestration. | **Advantages:**<br>▶ Comprehensive taxonomy of credit assignment and **CTDE** paradigms.<br>▶ Establishes formal theoretical criteria for communication protocols.<br>▶ Provides the theoretical foundation for PrismSpace's modular agent architecture.<br>**Disadvantages:**<br>▶ Survey paper without empirical experimental evaluation.<br>▶ Focuses on discrete game simulations rather than **API**-driven software tasks.<br>▶ Does not cover **LLM** reasoning agents or human-in-the-loop approval gates. |

# **HIAT** (Human-in-the-Loop with Auxiliary Tasks): Human-in-the-Loop **RL** with Auxiliary Tasks

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **B. Niu, B. Luo, Y. Cui, X. Xu, Y. Zhao, and Y. Feng**<br>*Artificial Intelligence Review*<br>**Year:** 2025<br>**Publisher:** Springer Open Access | Introduces **HIAT (Human-in-the-Loop with Auxiliary Tasks)**, a framework leveraging human intervention signals as auxiliary self-supervised learning objectives. Models human corrections as feedback gradients that shape internal feature representations, minimizing human intervention frequency over time. | **Advantages:**<br>▶ Maximizes learning efficiency from scarce human intervention signals.<br>▶ Significantly reduces human cognitive load through auxiliary reward shaping.<br>▶ Informs PrismSpace's approval gate telemetry and learning from user edits.<br>**Disadvantages:**<br>▶ Requires consistent human feedback to avoid biased auxiliary representations.<br>▶ Intervention timing latency can cause temporal mis-alignment in **RL** updates.<br>▶ Higher computational overhead during auxiliary task representation training. |

# **ORPO** (Odds Ratio Preference Optimization): Monolithic Preference Optimization without Reference Model

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **J. Hong, N. Lee, and J. Thorne** *EMNLP* / *arXiv:2403.07691* **Year:** 2024 **Area: LLM** Preference Alignment | **Core Idea (Single-Phase Alignment):** Instead of multi-step **RLHF**, **ORPO** teaches desired answers while penalizing bad answers in one unified equation: $$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda \cdot \mathcal{L}_{\text{OR}}$$ $$\mathcal{L}_{\text{OR}} = -\log \sigma \left( \log \frac{\text{odds}(y_w \mid x)}{\text{odds}(y_l \mid x)} \right)$$ **Key Terms:** ▶ $\mathcal{L}_{\text{SFT}}$: Base supervised fine-tuning loss (correct code/syntax). ▶ $\mathcal{L}_{\text{OR}}, \lambda$: Odds ratio penalty term and penalty weight. ▶ $x, y_w, y_l$: Prompt, winning action path, and losing action path. | **Advantages:** ▶ Reference-model-free: eliminates maintaining frozen $\pi_{\text{ref}}$ in memory. ▶ Monolithic single-phase training avoids unstable **PPO** policy optimization. ▶ Directly aligns PrismSpace agent trajectories on paired feedback (*chosen* vs. *rejected*). **Disadvantages:** ▶ Requires high-quality pairwise preference datasets $(y_w, y_l)$. ▶ Sensitive to $\lambda$ hyperparameter balance. |

# Reinforcement Online Learning to Rank with Unbiased Reward Shaping

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **S. Zhuang, Z. Qiao, and G. Zuccon**<br>*Information Retrieval Journal*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Develops an **Online Learning to Rank (OLTR)** framework using counterfactual reinforcement learning. Employs **inverse propensity weighting (IPW)** and unbiased reward shaping to correct for position, presentation, and click bias in user interaction logs during continuous ranker optimization. | **Advantages:**<br>▶ Guarantees unbiased gradient estimation from observational implicit feedback.<br>▶ Operates entirely online without requiring manual document relevance labels.<br>▶ Directly informs PrismSpace's context retrieval ranking feedback loop.<br>**Disadvantages:**<br>▶ High propensity score variance can destabilize gradient updates.<br>▶ Requires large volumes of query-click logs to converge.<br>▶ Assumes position-based user click models that may not hold in code search. |

# Knowledge Learning for Securing Workflow Scheduling in **MEC** (Mobile Edge Computing)

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **T. M. Ghazal, A. E. Awouda, M. K. Hasan, A. H. A. Rahman, S. Islam, R. A. Saeed, H. Elshafie, A. Iqbal, and A. H. Hussein**<br>*Mobile Networks and Applications*<br>**Year:** 2025<br>**Publisher:** Springer<br>Open Access | Proposes a **Knowledge-Guided Reinforcement Learning (KG-RL)** scheduler for complex workflow **DAGs** in distributed edge-cloud environments. Integrates security constraint validation directly into the state transition model, preventing insecure task placement while optimizing makespan. | **Advantages:**<br>▶ Enforces strict security boundary constraints during automated task scheduling.<br>▶ Effectively handles multi-dependency Directed Acyclic Graph (**DAG**) workflows.<br>▶ Informs PrismSpace's sandboxed tool execution and dependency resolution.<br>**Disadvantages:**<br>▶ Constraint checking introduces scheduling latency for massive **DAGs**.<br>▶ Assumes static security classification matrices across network nodes.<br>▶ Limited evaluation on non-deterministic tool failure scenarios. |

# Energy-Aware Cooperative Multi-Fitness Evolutionary Workflow Scheduling

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **P. Barredo and J. Puente**<br>*Natural Computing*<br>**Year:** 2025<br>**Publisher:** Springer<br>Open Access | Presents a **Cooperative Multi-Fitness Evolutionary Algorithm (CMFEA)** for multi-objective workflow scheduling. Optimizes trade-offs between execution time, monetary compute cost, and energy consumption using island-model cooperative coevolution across heterogeneous worker pools. | **Advantages:**<br>▶ Pareto-optimal trade-offs across makespan, financial cost, and energy.<br>▶ Cooperative sub-population decomposition prevents premature local convergence.<br>▶ Provides heuristics for PrismSpace's multi-agent model provider selection.<br>**Disadvantages:**<br>▶ Evolutionary search has higher runtime latency than single-pass ML heuristics.<br>▶ Cannot adapt instantaneously to real-time **API** rate limit drops.<br>▶ Requires parameter tuning across mutation, crossover, and migration rates. |

# Optimized Task Scheduling Using Hybrid **DL** (Deep Learning) and Metaheuristics

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **A. Heirati, M. Fartash, and M. Khalily-Dermany**<br>*Neural Processing Letters*<br>**Year:** 2026<br>**Publisher:** Springer Open Access | Combines **Deep Learning (LSTM (Long Short-Term Memory))** workload prediction with metaheuristic optimization (Harris Hawks Optimization) for proactive workflow dispatching. Predicts task execution durations and memory bottlenecks in advance to prevent node overloading and minimize queue waiting times. | **Advantages:**<br>▶ Proactive workload forecasting prevents sudden compute node thrashing.<br>▶ Hybrid pipeline balances search exploration with predictive dispatching.<br>▶ Relevant for predicting token latency and cost before agent invocation.<br>**Disadvantages:**<br>▶ Two-stage pipeline introduces inference overhead before scheduling decisions.<br>▶ **LSTM** prediction accuracy degrades under sudden irregular workload spikes.<br>▶ Complex offline model retraining requirements. |

# Cost-Aware Workflow Offloading in Edge-Cloud Computing via **GA** (Genetic Algorithm)

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **S. Abdi, M. Ashjaei, and S. Mubeen**<br>*The Journal of Supercomputing*<br>**Year:** 2024<br>**Publisher:** Springer Open Access | Proposes a **Genetic Algorithm-based workflow offloading** framework considering heterogeneous network bandwidth, deadline constraints, and monetary costs. Implements chromosome repair operators to preserve **DAG** precedence validity throughout mutation and crossover. | **Advantages:**<br>▶ Guarantees valid **DAG** dependency structures via specialized repair operators.<br>▶ Explicitly accounts for network transfer latency between pipeline stages.<br>▶ Informs subtask payload transfer between worker agents in PrismSpace.<br>**Disadvantages:**<br>▶ Computationally heavy for interactive real-time user query response.<br>▶ Assumes deterministic network transfer bandwidth in cloud environments.<br>▶ Does not model dynamic agent failures or retries. |

# Dynamic Role Discovery and Assignment in Multi-Agent Task Decomposition

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **Y. Xia, J. Zhu, and L. Zhu**<br>*Complex & Intelligent Systems*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Proposes **dynamic role discovery and assignment** that decomposes complex high-level objectives into lower-level subtasks. Uses learned persona embeddings and mutual information to adaptively allocate subtasks according to situational agent capabilities, eliminating rigid static assignments. | **Advantages:**<br>▶ Enables flexible subtask allocation without hardcoded role hierarchies.<br>▶ Hierarchical decomposition prevents single-agent cognitive context overload.<br>▶ Serves as core theoretical basis for PrismSpace's dynamic agent routing.<br>**Disadvantages:**<br>▶ Role discovery clustering introduces upfront computational overhead.<br>▶ Performance depends on representation vector accuracy for subtasks.<br>▶ Evaluated primarily on synthetic collaborative coordination benchmarks. |

# Decision Variable Contribution Based Adaptive Multi-Objective Scheduling

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **J. Li, L. Xing, W. Zhong, Z. Cai, and F. Hou**<br>*Complex & Intelligent Systems*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Introduces a **Decision Variable Contribution (DVC)** adaptive mechanism for evolutionary multi-objective workflow scheduling. Evaluates the differential sensitivity of individual decision variables to focus evolutionary search operators on high-impact scheduling bottlenecks. | **Advantages:**<br>▶ Accelerates Pareto-front convergence by prioritizing high-variance variables.<br>▶ Dynamically adapts search intensity across workflow bottleneck stages.<br>▶ Useful for pinpointing high-latency critical path tools in agent workflows.<br>**Disadvantages:**<br>▶ Sensitivity calculation adds per-generation matrix computation cost.<br>▶ Can over-concentrate on dominant objectives at the expense of diversity.<br>▶ Requires calibrated normalization across widely different metric units. |

# GHQ: Grouped Hybrid Q-Learning for Heterogeneous Multi-Agent RL

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **X. Yu, Y. Lin, X. Wang, S. Han, and K. Lv**<br>*Complex & Intelligent Systems*<br>**Year:** 2024<br>**Publisher:** Springer Open Access | Develops **Grouped Hybrid Q-learning (GHQ)** for cooperative heterogeneous multi-agent systems. Clusters agents with similar capabilities into functional cohorts that share value approximations while maintaining decoupled action execution heads for role specialization. | **Advantages:**<br>▶ Scales value estimation across large heterogeneous swarms via cohort grouping.<br>▶ Reduces joint action space dimensionality from exponential to block-linear.<br>▶ Directly applicable to categorizing **LLM** worker agent skill specializations.<br>**Disadvantages:**<br>▶ Cohort clustering boundaries can be rigid if agent capabilities evolve.<br>▶ Intra-group credit assignment remains non-trivial in dense interactions.<br>▶ Requires domain knowledge to establish initial grouping heuristics. |

# Decentralized Multi-Agent Cooperation via Adaptive Partner Modeling

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **C. Xu, J. Wang, X. Zhu, Y. Yue, W. Zhou, Z. Liang, and D. Wojtczak**<br>*Complex & Intelligent Systems*<br>**Year:** 2024<br>**Publisher:** Springer Open Access | Tackles environmental non-stationarity in decentralized multi-agent cooperation using **Adaptive Partner Modeling (APM)**. Instead of relying on expensive continuous inter-agent broadcast communication, agents infer partner policies and intended actions from local observation histories. | **Advantages:**<br>▶ Minimizes inter-agent communication bandwidth to near zero during execution.<br>▶ Stabilizes decentralized joint execution despite teammate policy shifts.<br>▶ Directly adopted in PrismSpace for zero-communication agent coordination.<br>**Disadvantages:**<br>▶ Partner inference error propagates if historical observations are noisy.<br>▶ Inference networks introduce local computational and memory overhead.<br>▶ Assumes partners follow relatively stationary parameterized policy spaces. |

# A Responsible AI Framework: Pipeline Contextualisation

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **E. Vyhmeister, G. G. Castañé, P.-O. Östberg, and S. Thevenin**<br>*AI and Ethics*<br>**Year:** 2023<br>**Publisher:** Springer<br>Open Access | Formulates an operational **Responsible AI Framework** providing pipeline contextualisation across the full model lifecycle. Specifies auditable checkpoints, data lineage tracking, and bias mitigation protocols for autonomous AI decision-making systems. | **Advantages:**<br>▶ Holistic governance framework linking technical architecture to ethical safety.<br>▶ Defines concrete telemetry points for data provenance and action auditing.<br>▶ Guides PrismSpace's execution audit logging and human inspection UI.<br>**Disadvantages:**<br>▶ Qualitative framework that requires custom translation into concrete code gates.<br>▶ Audit logging introduces storage overhead and slight execution latency.<br>▶ Does not provide automated formal verification of **LLM** reasoning traces. |

# Meaningful Human Control: Actionable Properties for AI Development

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **L. Cavalcante Siebert, M. L. Lupetti, E. Aizenberg, N. Beckers, A. Zgonnikov, H. Veluwenkamp, D. Abbink, E. Giaccardi, G.-J. Houben, C. M. Jonker, J. van den Hoven, D. Forster, and R. L. Lagendijk**<br>*AI and Ethics*<br>**Year:** 2023<br>**Publisher:** Springer<br>Open Access | Formalizes the philosophical concept of **Meaningful Human Control (MHC)** into actionable software design properties: **Tracking** (system responsive to human moral reasons) and **Tracing** (system actions causally attributable to accountable human supervisors). | **Advantages:**<br>▶ Translates abstract governance principles into rigorous software specifications.<br>▶ Establishes tracking and tracing requirements for autonomous tool execution.<br>▶ Directly guides the design of PrismSpace's Human-in-the-Loop (**HITL**) approval gates.<br>**Disadvantages:**<br>▶ High human intervention frequency can degrade automated agent throughput.<br>▶ Defining actionable reason boundaries requires careful domain calibration.<br>▶ Assumes human supervisor maintains full comprehension of complex agent plans. |

# **CPN-HRL** (Cloud-Private-Network Hierarchical Reinforcement Learning): Hierarchical **DRL** for Priority-Aware Task Scheduling

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **Q. Yue, L. Tian, X. Feng, Z. Yuan, S. Wei, J. Xia, B. Chen, X. Song, and Y. Yao**<br>*Journal of King Saud University CIS*<br>**Year:** 2026<br>**Publisher:** Springer Open Access | Presents **CPN-HRL**, a Hierarchical Reinforcement Learning framework for priority-aware task scheduling in Cloud-Edge-End networks. Uses a macro-policy to partition task **DAGs** by priority and latency budgets, while micro-policies schedule subtasks onto optimal compute nodes. | **Advantages:**<br>▶ Hierarchical policy decomposition effectively handles complex task **DAGs**.<br>▶ Guarantees priority-aware quality of service under tight latency deadlines.<br>▶ Serves as primary reference for PrismSpace's **DAG** workflow scheduling engine.<br>**Disadvantages:**<br>▶ Multi-tier policy architecture requires synchronous training coordination.<br>▶ Edge node failure requires immediate micro-policy replanning.<br>▶ Evaluated in simulated telecom cloud-edge topologies. |

# Automatic Discovery and Description of Human Planning Strategies

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **J. Skirzyński, Y. Jain, and F. Lieder** *Behavior Research Methods* **Year:** 2023 **Publisher:** Springer Open Access | Develops an algorithmic framework for **automatically discovering and formalizing planning strategies** from human behavior traces. Models task planning as resource-rational search over Directed Graphical Models, producing interpretable decision trees that characterize human heuristic planning. | **Advantages:**<br>▶ Extracts interpretable, human-like heuristic planning strategies automatically.<br>▶ Quantifies trade-offs between cognitive search effort and decision quality.<br>▶ Informs PrismSpace's hierarchical prompt decomposition and agent planning.<br>**Disadvantages:**<br>▶ Search space for strategy discovery grows exponentially with problem depth.<br>▶ Human behavioral experiments conducted in stylized cognitive testbeds.<br>▶ Static decision trees struggle to capture fluid **LLM** reasoning trajectories. |

# Data-Intensive Workflow Scheduling Based on **DRL** in Multi-Clouds

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **S. Zhang, Z. Zhao, C. Liu, and S. Qin**<br>*Journal of Cloud Computing*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Proposes a **Deep Q-Network (DQN)** scheduler optimized for data-intensive workflows across multi-cloud environments. Explicitly models dataset placement and cross-datacenter transfer latency to minimize both workflow completion time and data egress billing. | **Advantages:**<br>▶ Optimizes data locality to prevent massive cross-cloud egress overhead.<br>▶ Adaptive reward function balances monetary cost and makespan constraints.<br>▶ Directly relevant to caching and embedding retrieval across distributed agents.<br>**Disadvantages:**<br>▶ Assumes deterministic file sizes and network bandwidth availability.<br>▶ Q-network value overestimation requires double-**DQN** correction.<br>▶ State representation size scales with the total number of cloud datacenters. |

# **RL**-Based Task Scheduling for Sustainable Federated Cloud Computing

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **Z. Wang, S. Chen, L. Bai, J. Gao, J. Tao, R. R. Bond, and M. D. Mulvenna**<br>*Journal of Cloud Computing*<br>**Year:** 2023<br>**Publisher:** Springer Open Access | Develops an **Actor-Critic RL scheduling framework** for federated cloud datacenters that incorporates renewable energy availability into scheduling policies. Dynamically routes compute tasks to datacenters with green energy surpluses without violating user SLA deadlines. | **Advantages:**<br>▶ Incorporates environmental sustainability and carbon metrics into scheduling.<br>▶ Actor-Critic policy gradient handles continuous action spaces effectively.<br>▶ Provides conceptual foundation for multi-attribute provider routing.<br>**Disadvantages:**<br>▶ Requires real-time telemetry from green energy generation sources.<br>▶ Trade-off between carbon footprint and network transmission latency.<br>▶ Sensitive to reward scaling factors between SLA penalties and green rewards. |

# Hybrid Dolphin Swarm Sparrow Search Multi-Objective Workflow Scheduling

| Citation | Summary | Advantages & Disadvantages |
|---|---|---|
| **S. Subashree, M. Rajakumaran, G. Pushpa, and S. Senthilkumar**<br>*Journal of Cloud Computing*<br>**Year:** 2025<br>**Publisher:** Springer Open Access | Proposes a hybrid bio-inspired metaheuristic combining **Dolphin Swarm Optimization (DSA)** and **Sparrow Search Algorithm (SSA)** for multi-objective cloud workflow scheduling. Optimizes execution makespan, virtual machine utilization, and monetary execution expenditure. | **Advantages:**<br>▶ Combines global exploration of DSA with local exploitation of SSA.<br>▶ Demonstrates superior Pareto hypervolume compared to standard **GA** and **PSO** (Particle Swarm Optimization).<br>▶ Provides strong comparative baselines for workflow optimization algorithms.<br>**Disadvantages:**<br>▶ Metaheuristic iteration runtime is too slow for sub-second reactive routing.<br>▶ Multiple tuning parameters (predation rate, vocalization index) required.<br>▶ Lacks theoretical convergence rate bounds for stochastic optimization. |

# Research Gap

## What the surveyed methods already do

**MARL** works address **CTDE**, role grouping, and ad-hoc coordination. **ORPO** aligns **LLM** preferences without reference models, while **HIAT** leverages human interventions as auxiliary **RL** signals. **CPN-HRL** and **DAG** schedulers optimize cloud task allocation, and **MHC** formalizes tracking and tracing properties.

## Limitations observed in those papers

Coordination benchmarks focus on gaming (**SMAC**, Overcooked) rather than filesystem/**API** tooling. **ORPO** lacks grounding in **MCP ReAct** tool traces, while **HIAT** targets continuous robotic control instead of code execution. Cloud schedulers assume static **DAGs** or slow metaheuristics ill-suited for real-time **LLM** routing.

## Why PrismSpace is required

No reviewed system jointly grounds tool observations, aligns policies on agent trajectories, dynamically routes heterogeneous **LLMs**, and enforces human-in-the-loop safety gates within a unified developer workspace.

# 03. Comparative Analysis: Existing Platforms vs. PrismSpace

**Simple comparison:** Comparing PrismSpace with conversational AI, autonomous-agent, and multi-agent platforms.

| Capability Dimension | ChatGPT Web (OpenAI) | AutoGPT (Autonomous Loop) | CrewAI (Multi-Agent Swarm) | PrismSpace (Ours) Local Multi-Agent Platform |
|---|---|---|---|---|
| Execution Topology | One conversation at a time | Repeats steps automatically; may not stop | Agents follow a fixed sequence | **DAG (Directed Acyclic Graph) Scheduling:** Runs dependent tasks in parallel, with up to three workers |
| Model Strategy & Routing | Mostly one provider and model | Usually one model with custom scripts | Model selected for each agent role | **Multi-Model Router:** Selects a suitable provider using task needs, speed, and cost |
| Tool Connectivity Protocol | Built-in plugins and function calls | Custom scripts and shell commands | Tool wrappers with fixed formats | **MCP (Model Context Protocol):** A common interface for tools such as Gmail, Drive, files, and web services |
| Observability & Streaming | Shows answer text; internal steps are hidden | Terminal logs only | Text traces from agent execution | **Live Run Status + Workflow Graph:** Shows task progress and each node's state |
| Safety & Quality Control | Provider filters; no user approval step | Fixed limits; limited output checking | Agent reviews its own output | **HITL (Human-in-the-Loop) + ORPO (Odds Ratio Preference Optimization):** Pauses risky tasks and ranks preferred outputs |
| Persistence & Privacy | Data stored in the provider's cloud | Local files without a structured database | Cloud or temporary in-memory state | **Local-First PrismDB:** Stores sessions and files in the browser's IndexedDB database |

# Project Architecture, Working & System Design

Project workflow, architecture, working process, agent lifecycle, dataset pipeline, DAG scheduling, and flow diagrams

# 06. PrismSpace Workflow (Figure 1.0)

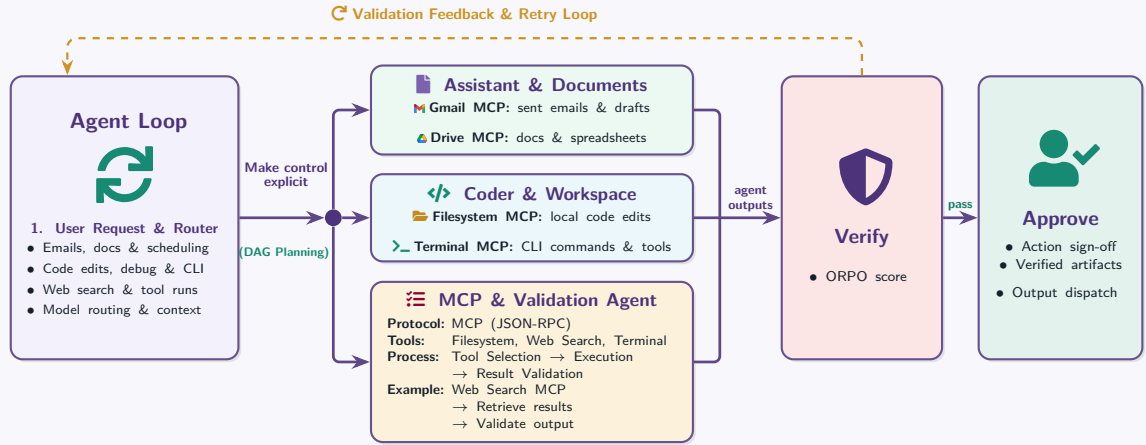



**Figure 1.0: PrismSpace Workflow** | **DAG** = Directed Acyclic Graph | **MCP** = Model Context Protocol | **ReAct** = Reason + Act | **ORPO** = Odds Ratio Preference Optimization | **HITL** = Human-in-the-Loop

# 10. Model Training Map

## Model-to-Algorithm Training Matrix

| Model | Type | ML algorithm for training | Why choose this algorithm? |
|---|---|---|---|
| **Intent Classifier + FAISS Retrieval Index** | Classification + unsupervised retrieval | XGBoost classifier + sentence embeddings / FAISS | Classifies the user's intent and retrieves semantically relevant research or tool context for the request. |
| **Agent Router** | Multilabel routing | Binary-relevance XGBoost | Assigns one or more specialist agents independently from task, capability, and tool-use features. |
| **Provider Router** | Classification / ranking | XGBoost classifier + ranking | Filters eligible providers and ranks the best model using capability, quality, latency, cost, and availability. |
| **Reward ORPO Adapter** | ORPO | ORPO + Reinforcement Learning | **ORPO (Odds Ratio Preference Optimization):** aligns language models using preference data. It is a reward-model-free alternative that eliminates the need for a separate reward model or reference model. |

# 09. Model Responsibilities & Inference Flow



# Intent Classifier: Prompt Decomposition, NLP Algorithm & Serialization

## 1 User Input

**User**
(Human)

*"Find the latest API changes, summarize them, and update my notes."*

**Function**

Receive the user request and prepare it for processing.

**Input**

Natural language prompt (from user).

**Output**

Raw prompt text.

## 2 Tokenize / Encode

**Tokenizer Agent**
(Tokenizer Service)

**Function**

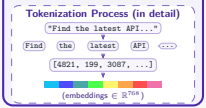
Tokenize prompt into IDs and compute dense embeddings.

**Input**

Raw prompt text.

**Output**

Token IDs + embedding vectors.




## 3 Intent Classifier

**Intent Classifier Agent**
(LLM / Classifier)

**Function**

Analyze contextual embeddings to classify user intent, extract domain entities (target API, destination notes), resolve dependencies, and evaluate confidence thresholds for task routing.

**Input**

Token embeddings + prompt context.

**Output**

Intent label(s) with confidence (research: 0.98, summarize: 0.95, write: 0.92) + entity slots.

## 4 Semantic Decomposer

**Decomposer Agent**
(Planning Agent)

**Function**

Break the classified intent into modular sub-tasks, establish execution ordering, and detect dependencies.

**Input**

Intent labels + context from Step 3.

**Output**

JSON-RPC task packet:

```
{"tasks": ["research",
"extract", "summarize",
"write"],
"approval_gate": true
}
```

## 5 Task DAG Output

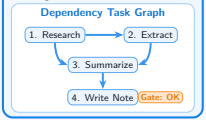
**Tool Executor Agent**
(Tooling Agent)

**Function**

Construct, validate, and execute the Directed Acyclic Graph (DAG) for dependency-ordered tool execution.

**Input**

Serialized task packet + tool bindings.




## Tokenization Process (Step 2 – Detailed)

### 1. Raw Prompt

**Input**

"Find the latest API changes, summarize them, and update my notes."

*(string)*

### 2. Tokenization

**Function**

Split text into tokens (using BPE / WordPiece).

**Output**

`["Find", "the", "latest", "API", "changes", ...]`

*(tokens)*

### 3. Convert to IDs

**Function**

Map tokens to unique integers from vocabulary.

**Output**

`[4821, 199, 3087, 421, ...]`

*(token IDs)*

### 4. Create Embeddings

**Function**

Convert IDs to vector representations (e.g., 768-d).

**Output**



`[0.12, 0.45, -0.03, ...]`

*(embedding vectors)*

### 5. Pass to Intent Classifier

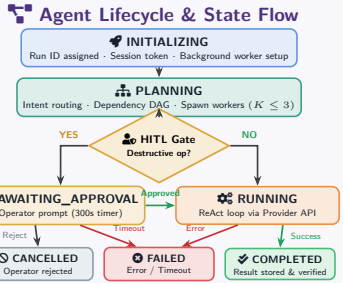
**Function**

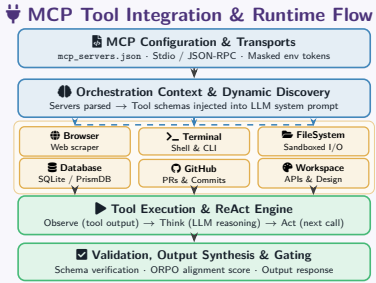
Embeddings + context → intent classification.

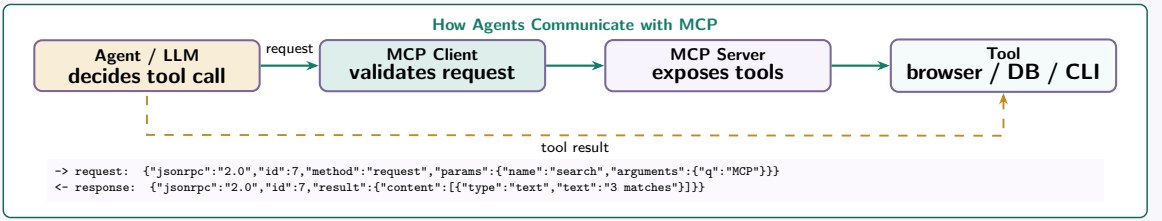
**Output**

Intent label(s) + entities ("research", "summarize", ...).

# 07. Agent Lifecycle, MCP Integration & Runtime Flow in PrismSpace



# 11. How the System Works: From Request to Result

## Stage 1: Ingestion & Features

- **Capture request:** Parse user input (text/voice/API).
- **Clean & normalize:** Remove noise, handle missing data with NLP preprocessing.
- **Extract features:** Tokenize into semantic embeddings using transformer models.

## Stage 2: Intelligence & DAG

- **Classify intent:** ML classifier identifies task type from 12 categories (99.6% accuracy).
- **Choose agents & models:** Route sub-tasks to specialized LLM agents.
- **Topological sorting:** Order DAG tasks by dependencies; execute only after all prerequisites finish.

## Stage 3: Execution & Storage

- **Execute:** Run up to 3 parallel tasks using Model Context Protocol (MCP) tools.
- **Track progress:** Real-time telemetry, node state transitions, and execution logs.
- **Verify & store:** ORPO quality check, persist results to PrismDB.

## Building & Executing the Task Graph $\mathcal{G}(V, E)$

**1. User Goal Prompt**
"Analyze news+CSV, email summary"

→

**2. Split into Vertices $V$**
Modular async tasks $A$, $B$, $C$, $D$

→

**3. Connect Edges $E$**
Map output return → input args

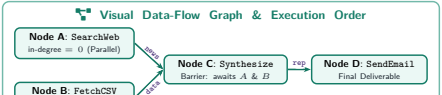
→

**4. Topological Sort**
Pick in-degree $= 0$; unlock dependents

### Computer Science Explanation (By Concrete Example):

- **Vertices ($V$) = Sub-Task Functions (Async Call Units):**
  Node $A$: `news = await SearchWeb("Tesla news")` *[Researcher]*
  Node $B$: `data = await FetchCSV("metrics.csv")` *[Data Agent]*
  Node $C$: `rep = await Synthesize(news, data)` *[Writer Agent]*
  Node $D$: `await SendEmail(to="lead", body=rep)` *[Email Tool]*
- **Edges ($E$) = Data Pipes & Synchronization Barriers:**
  Edge $(A, C)$ pipes return value `news`; edge $(B, C)$ pipes `data`.
  Barrier: task $C$ blocks until both promises $A$ & $B$ resolve.
- **Connection Basis:** $C$ links to $A$ & $B$ because its inputs require their outputs; $A$ and $B$ are independent.
- **Topological Sorting (Kahn's method):** repeatedly select nodes with in-degree $= 0$, remove outgoing edges, and unlock dependents.
  This gives a valid dependency order and exposes independent work for parallel execution.
- **Execution:** $A$ and $B$ run in parallel; after both finish, $C$ unlocks, then $D$ runs.

### Visual Data-Flow Graph & Execution Order




**Parallel Thread 1:** Worker 1 executes Node A → returns `news`
**Parallel Thread 2:** Worker 2 executes Node B → returns `data`
**Join Barrier:** Node C consumes `news` + `data` → outputs `rep`

**Serial Finish:** Node D consumes `rep` → sends email

# Why Topological Sorting is Chosen for DAG Task Scheduling

### 1. Task Dependencies

- **Prerequisites first:** A task starts after the tasks it needs are finished.
- **Independent work:** Unrelated tasks can run in any order.

### 2. Finds Waiting Loops

- **Finds circular waiting:** Detects loops such as $A$ waits for $B$, while $B$ waits for $A$.
- **Stops safely:** Reports the problem instead of waiting forever.

### 3. Runs Work in Parallel

- **Ready tasks:** Tasks with no unfinished prerequisites can start together.
- **Safe hand-off:** The next task starts only when all required inputs arrive.

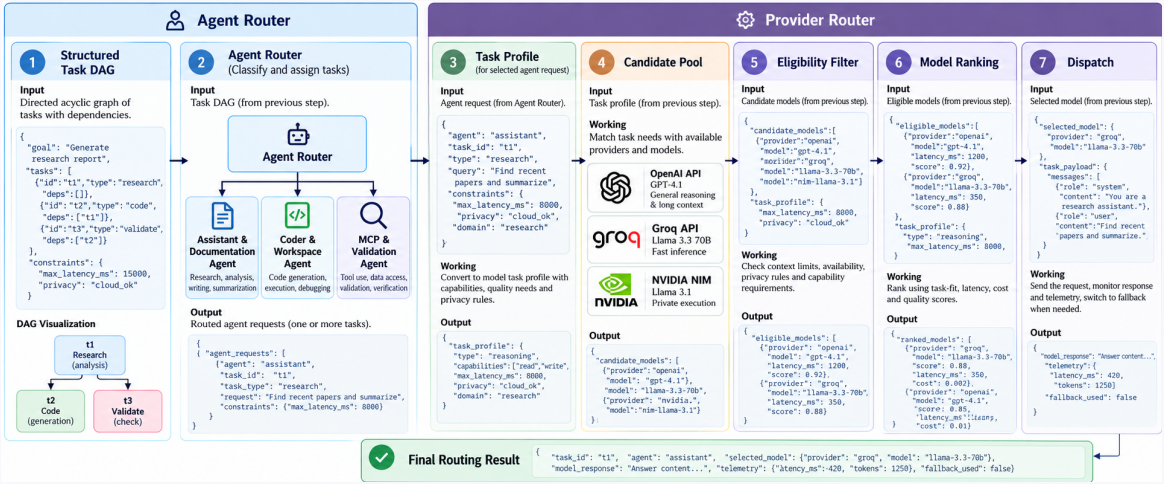
### Simple Comparison: Topological Sort vs. Common Sorting

| Method | Works With | How It Orders | Speed | Finds Loops | Parallel Work | Useful for Task Workflows |
|---|---|---|---|---|---|---|
| **Topological Sort** | Task dependency graph | **Prerequisites first** | $\mathcal{O}(V + E)$ | Yes (In-degree) | Yes (Fan-out) | **Best choice:** Follows task dependencies and runs independent tasks together. |
| **Merge Sort** | List of values | **Smallest value first** | $\mathcal{O}(n \log n)$ | No (N/A) | No (Serial 1D) | **Not suitable:** Sorts values, not task prerequisites. |
| **Quick Sort** | List of values | **Around a pivot value** | $\mathcal{O}(n \log n)$ avg | No (N/A) | No (Serial 1D) | **Not suitable:** Does not know which task must come first. |
| **Heap Sort** | Priority list | **Highest priority first** | $\mathcal{O}(n \log n)$ | No (N/A) | No (Serial 1D) | **Not suitable:** Priority alone cannot enforce several prerequisites. |
| **Bubble / Insertion** | List of values | **Neighbor comparisons** | $\mathcal{O}(n^2)$ | No (N/A) | No (Strict Serial) | **Not suitable:** Pairwise swaps cannot represent task dependencies. |

**Why This Matters for PrismSpace:** Regular sorting puts numbers or words in order. PrismSpace must order tasks based on their prerequisites. **Topological sorting** follows those prerequisites, detects circular dependencies, and lets independent tasks run together.

# Agent & Provider Router working

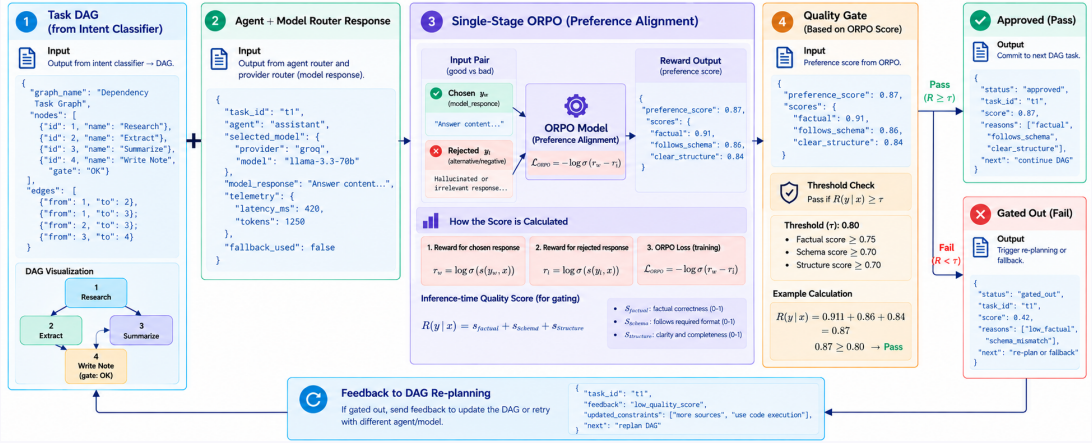
**Routing goal:** Convert a task profile into one suitable model request, then return the model response.



# How ORPO Works: Preference Alignment and Quality Gating

**What is ORPO & Why is it Used Here as a Reward System?**

- **What is ORPO? Odds Ratio Preference Optimization (ORPO)** is an integrated **single-stage reward and alignment algorithm**. Traditional **Reinforcement Learning from Human Feedback (RLHF)** requires training a separate reward model and keeping a frozen reference model in GPU memory (heavy, slow, and unstable).
- **Why in PrismSpace?** In multi-agent swarms, LLMs can hallucinate invalid tools or malformed code. ORPO acts as an **internal reward filter**: it mathematically rewards compliant trajectories ($y_w$) and penalizes bad responses ($y_l$) during fine-tuning, eliminating the need for an external reward model.



# Algorithm

Step-by-step execution of the PrismSpace multi-agent workflow

# Algorithm: Step 1 – Understand the User Prompt

## Step 1: Prompt Parsing and Structured Representation

**1.1** Ingest raw natural language prompt $\mathcal{T}_{\text{user}}$ via client interface and extract core objective.

**1.2** Retrieve persistent session context and workspace history from browser IndexedDB (`PrismDB`):

- **1.2.1** Load prior interaction turns, agent memory checkpoints, and active user configurations.
- **1.2.2** Resolve local file system references, target project repositories, and access tokens.

**1.3** Construct strongly-typed operational execution tuple:

$$T = \langle \text{Goal, Constraints, Context, } \mathcal{M}_{\text{tools}}, \Phi_{\text{out}} \rangle$$

**Where:** Goal: Target objective | Constraints: Safety & SLA bounds | Context: Session state in `PrismDB` | $\mathcal{M}_{\text{tools}}$: Authorized MCP tools | $\Phi_{\text{out}}$: Deliverable schema contract.

- **1.3.1** Parse explicit operational constraints (deadlines, schema specifications, output formats).
- **1.3.2** Bind authorized Model Context Protocol (MCP) tool capabilities and environment permissions.

**1.4** Perform input boundary and security screening:

- **1.4.1** Evaluate prompt against token limits and scan for prompt injection patterns.
- **1.4.2** Flag state-mutating actions requiring mandatory human-in-the-loop authorization.

## Step 2: Intent Classification and Capability Check

**2.1** Categorize user prompt into functional operational domains:

$$\text{Domain} \in \{\text{Productivity/Automation, Document Synthesis, Web Research, Code Engineering}\}$$

**Where:** Productivity: Communication & planning | Synthesis: Summarization | Research: Web search & scraping | Coding: CLI & code synthesis.

- **2.1.1** Compute dense semantic embeddings and syntactic keyword vectors for input prompt.
- **2.1.2** Route through multi-class classifier head $\hat{\mathbf{y}}_{\text{intent}}$ to determine target capability domain.

**2.2** Profile required system capabilities and operational risk levels:

- **2.2.1** Determine required subsystem resources (live web search, local filesystem I/O, terminal execution).
- **2.2.2** Enforce safety categorization: Read-Only (autonomous execution) vs. State-Mutating (approval-gated).

**2.3** Select optimal execution pathway:

- **2.3.1** Route single-step queries to direct, low-latency single-model inference pipeline.
- **2.3.2** Dispatch multi-step complex requests to DAG Planner agent for task decomposition.

## Step 3: DAG Decomposition and Model Routing

**3.1** Deconstruct objective into Directed Acyclic Graph $\mathcal{G} = (V, E)$:

- **3.1.1** Instantiate discrete task vertices $v_i \in V$ with atomic execution contracts and output targets.
- **3.1.2** **Data Dependencies as Directed Edges** ($e = \langle v_i, v_j \rangle \in E \iff \mathcal{O}_i \cap \mathcal{I}_j \neq \emptyset$): Task $v_j$ requires input artifacts produced by $v_i$ (e.g., synthesis requires scraped research).
- **3.1.3** **Execution Barrier:** Consumer $v_j$ remains blocked until producer $v_i$ delivers $\mathcal{O}_i$, guaranteeing tasks never run on missing or partial data.

**3.2** Compute topological task execution schedule $\pi = \mathrm{TopologicalSort}(\mathcal{G})$:

- **3.2.1** Identify independent vertices (in-degree $= 0$) for concurrent parallel dispatch ($K \leq 3$).
- **3.2.2** Schedule downstream vertices dynamically upon upstream output event emission.

**3.3** Solve multi-objective provider routing optimization for each sub-task $v_i$:

$$\text{Provider}^* = \arg\min_m \left( \lambda_{\text{cost}} \, \text{Cost}_m + \lambda_{\text{latency}} \, \text{Latency}_m \right) \quad \text{s.t. } \text{Quality}_m \geq \tau$$

**Where:** Provider$^*$: Optimal LLM backend | $m$: Candidate model | $\text{Cost}_m, \text{Latency}_m$: Normalized price & delay | $\lambda_{\text{cost}}, \lambda_{\text{latency}}$: Balancing weights | $\tau$: Quality barrier.

- **3.3.1** Route high-throughput, latency-critical nodes to LPU backends (Groq / Llama-3).
- **3.3.2** Route deep reasoning and code generation tasks to frontier backends (OpenAI GPT-4o, Claude 3.5).

## Step 4: Role-Based Task Assignment and Dispatching

**4.1** Query specialist agent registry and capability profiles:

$$\mathcal{A} = \{\text{Planner, Assistant, Researcher, Coder, Validator}\}$$

**4.1.1** Evaluate persona system prompts, specialized instruction boundaries, and assigned MCP tools.

**4.1.2** Verify agent worker availability and monitor concurrent thread utilization.

**4.2** Execute capability matching algorithm for each sub-task node $v_i$:

$$\text{Agent}^* = \arg\max_{a \in \mathcal{A}} \ \text{CosineSimilarity}\big(\mathbf{Req}(v_i),\, \mathbf{Cap}(a)\big)$$

**Where:** Agent$^*$: Assigned specialist persona | $\mathcal{A}$: Swarm agent registry | $\mathbf{Req}(v_i)$: Task requirement vector | $\mathbf{Cap}(a)$: Agent capability embedding | $\text{CosineSimilarity}$: Semantic alignment.

**4.2.1** Match information gathering requirements (Tag $\ni$ `web_search`) to Researcher agent.

**4.2.2** Match software synthesis and execution (Tag $\ni$ `python, cli`) to Coder agent.

**4.2.3** Match quality assurance and policy compliance (Tag $\ni$ `audit, verify`) to Validator agent.

**4.3** Manage task lifecycle states in dispatcher queue:

**4.3.1** Place sub-tasks in `PENDING` state until all parent dependency inputs resolve.

**4.3.2** Transition verified nodes to `READY` and dispatch execution payloads to assigned agents.

## Step 5: Agent Decision Cycle and Inter-Agent Communication

**5.1** Execute autonomous Reasoning-Action (ReAct) cycle for each dispatched agent:

$$\text{State } s_t \longrightarrow \textbf{Observe} \longrightarrow \textbf{Reason} \longrightarrow \textbf{Act} \longrightarrow \textbf{Reflect}$$

**Where:** $s_t$: Current execution state at turn $t$ | **Observe**: Ingest context & parent outputs $\mathcal{O}_i$ | **Reason**: Plan via CoT | **Act**: Invoke MCP tools | **Reflect**: Self-correct on error.

- **5.1.1** **Observe:** Ingest assigned sub-task description, operational constraints, and parent inputs.
- **5.1.2** **Reason:** Formulate intermediate chain-of-thought rationale to plan next tool invocation.
- **5.1.3** **Act:** Dispatch structured JSON-RPC tool calls to external MCP servers or formulate text.
- **5.1.4** **Reflect:** Inspect execution exit codes and output schemas; initiate local correction on errors.

**5.2** Transmit intermediate artifacts via DAG edge message passing:

- **5.2.1** Route validated producer artifact $\mathcal{O}_i$ directly into consumer input buffer $\mathcal{I}_j$.
- **5.2.2** Enforce strict artifact isolation to prevent cross-task context contamination and token inflation.

**5.3** Synchronize shared state via local blackboard memory:

- **5.3.1** Commit global project state, conversation turns, and file handles to IndexedDB (`PrismDB`).
- **5.3.2** Dispatch event-driven signals upon node resolution to awaken dependent downstream workers.

## Step 6: Specialist Execution Workflows – Planner & Assistant

**6.1** Execute Planner Agent workflow (System Architect & Coordinator):

- **6.1.1** Ingest user objective string and conversation context from `PrismDB`.
- **6.1.2** Decompose objective into atomic independent and dependent sub-goals.
- **6.1.3** Construct DAG graph vertices $V$, dependency edges $E$, and capability tags.
- **6.1.4** Solve provider routing optimization and enqueue topological schedule into Dispatcher.

**6.2** Execute Assistant Agent workflow (Productivity & Communication):

- **6.2.1** Retrieve user communication preferences, profiles, and reference documents from storage.
- **6.2.2** Formulate structured response drafts, templates, or executive summaries.
- **6.2.3** Interface with Google Workspace MCP tools (Gmail draft creation, Drive document retrieval).
- **6.2.4** Emit verified communication draft; enforce approval gate prior to external message transmission.

## Step 7: Specialist Execution Workflows – Researcher, Coder & Validator

**7.1** Execute Researcher Agent workflow (Information Retrieval & Synthesis):

- **7.1.1** Formulate optimized web search queries and invoke Browser MCP server endpoints.
- **7.1.2** Scrape target web documents, parse DOM trees, and extract relevant contextual passages.
- **7.1.3** Deduplicate findings and compile structured Markdown report with verified source citations.

**7.2** Execute Coder Agent workflow (Software Synthesis & Execution):

- **7.2.1** Inspect local repository tree via Filesystem MCP and synthesize target source code files.
- **7.2.2** Execute scripts inside Sandboxed Terminal CLI and inspect exit codes and error streams.
- **7.2.3** Analyze stack traces and compiler lints to autonomously self-correct syntax and runtime bugs.

**7.3** Execute Validator Agent workflow (Quality Auditing & Safety Gate):

- **7.3.1** Scan candidate outputs against safety policies, token invariants, and functional constraints.
- **7.3.2** Emit binary verification signal: **PASS** to release artifact, or **FAIL** with error diagnostics.

## Step 8: Standardized Tool Execution via Model Context Protocol

**8.1** Interface agents to external tools via Model Context Protocol (MCP):

- **8.1.1** Standardize client-server communication using JSON-RPC protocol specifications.
- **8.1.2** Enable dynamic tool discovery: inspect exposed tool schemas, input parameters, and return types.
- **8.1.3** Decouple agent reasoning logic from external tool implementation details.

**8.2** Orchestrate multi-domain tool execution ecosystem:

- **8.2.1** **System & Dev Tools:** Local Filesystem (file I/O), Sandboxed CLI (isolated shell), GitHub API.
- **8.2.2** **Productivity Tools:** Google Workspace MCP (Gmail messaging, Google Drive document access).
- **8.2.3** **Web Intelligence:** Browser MCP for real-time document search, retrieval, and scraping.

**8.3** Enforce Human-in-the-Loop (HITL) authorization policy:

- **8.3.1** Permit autonomous execution for non-destructive, read-only tool queries.
- **8.3.2** Intercept state-mutating actions (file modification, email dispatch, CLI execution) for user approval.

## Step 9: Context Compaction and Shared Memory Management

**9.1** Address context window constraints and token consumption limits:

**9.1.1** Mitigate fixed LLM context window boundaries to eliminate token overflow and latency degradation.

**9.1.2** Prevent quadratic attention computation overhead across long multi-agent interaction sessions.

**9.2** Execute dynamic context compaction and artifact pruning:

$$C_{t+1} = \mathrm{Prune}\big(C_t \cup \Delta\text{Artifacts} \setminus \text{Stale Tokens}\big)$$

**Where:** $C_{t+1}, C_t$: Next and current context buffers | $\Delta$Artifacts: Verified upstream deliverables $\mathcal{O}_i$ | Stale Tokens: Discarded scratchpad tokens & raw tool logs | $\mathrm{Prune}(\cdot)$: Heuristic eviction filter.

**9.2.1** Forward only verified parent output artifacts $\mathcal{O}_i$ to dependent downstream child nodes.

**9.2.2** Evict intermediate reasoning scratchpads, redundant tool telemetry, and discarded draft tokens.

**9.3** Persist global execution state using client-side database (`PrismDB`):

**9.3.1** Commit session turns, task graphs, and tool execution logs to browser IndexedDB via Dexie.js.

**9.3.2** Ensure client data privacy, full offline execution persistence, and instant crash recovery.

## Step 10: Single-Stage Preference Alignment (ORPO) and Verification

**10.1** Single-stage reference-free preference alignment formulation (ORPO):

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{OR}} \cdot \mathcal{L}_{\text{OR}} \quad \text{where} \quad \mathcal{L}_{\text{OR}} = -\log \sigma \left( \log \left[ \frac{\text{odds}_\theta(y_w \mid x)}{\text{odds}_\theta(y_l \mid x)} \right] \right)$$

**Where:** $\mathcal{L}_{\text{ORPO}}$: Total loss | $\mathcal{L}_{\text{SFT}}$: SFT cross-entropy loss | $\lambda_{\text{OR}}$: Odds weight | $\mathcal{L}_{\text{OR}}$: Odds ratio penalty | $\text{odds}_\theta(y \mid x) = \frac{P_\theta(y|x)}{1-P_\theta(y|x)}$: Output sequence odds | $y_w, y_l$: Chosen / rejected responses for prompt $x$.

**10.1.1** Jointly maximize chosen likelihood while penalizing rejected tokens ($2\times$ memory savings).

**10.1.2** Supervise via UltraFeedback pairs $\mathcal{D} = \{(x, y_w, y_l)\}$, bypassing unstable RLHF loops.

**10.2** Enforce runtime quality gating via preference score barrier $\mathcal{S}_{\text{ORPO}}(x, y) \geq \tau_{\text{approval}}$:

**10.2.1** Audit deliverables using fine-tuned model (`reward_orpo_v2`, Qwen2.5-1.5B LoRA).

**10.2.2** Score candidate outputs on instruction adherence, structural constraints, and safety guardrails.

**10.3** Handle quality failures via targeted DAG re-planning:

**10.3.1** Capture diagnostic error traces and schema deviations when output falls below $\tau_{\text{approval}}$.

**10.3.2** Dispatch error logs to DAG Planner for targeted branch re-planning without system abort.

## Step 11: Artifact Assembly and Global State Commit

**11.1** Aggregate and assemble multi-agent deliverables:

**11.1.1** Harvest verified partial output artifacts $\mathcal{O}_i$ from all terminal nodes in the completed DAG.

**11.1.2** Coalesce research summaries, verified code diffs, and formatted text into unified final deliverable.

**11.2** Calculate end-to-end composite delivery quality metric:

$$\text{Delivery Quality } D = \text{Correctness} \times \text{Completeness} \times \text{Timeliness}$$

**Where:** $D \in [0, 1]$: Composite quality metric | Correctness: Schema conformance & execution pass rate | Completeness: Fulfillment of goal constraints | Timeliness: Latency penalty score.

**11.2.1** Apply multiplicative scoring penalty: failure in any single dimension sharply penalizes overall quality.

**11.2.2** Log quality vector to persistent performance ledger to calibrate future provider routing models.

**11.3** Commit finalized session state with ACID guarantees:

**11.3.1** Atomically write completed conversation records, DAG execution graphs, and tool receipts to `PrismDB`.

**11.3.2** Record end-to-end telemetry: aggregate latency, per-agent token consumption, and audit receipts.

## Step 12: Output Delivery and Reactive Workspace Presentation

**12.1** Synthesize and package final deliverable artifacts:

- **12.1.1** Compile structured digital artifacts (Markdown reports, verified code patches, email/document drafts).
- **12.1.2** Generate comprehensive audit receipt containing agent provenance, execution duration, and tool logs.

**12.2** Render presentation layer via multi-pane workspace interface:

- **12.2.1** Update interactive DAG HUD with real-time node states (`Queued` → `Running` → `Success` / `Failed`).
- **12.2.2** Mount Artifact Viewer canvas providing rich Markdown rendering, syntax highlighting, and export controls.
- **12.2.3** Present interactive approval cards for pending external actions with explicit authorization controls.

**12.3** Synchronize client state reactively via local database transactions:

- **12.3.1** Commit verified state transitions and output artifacts directly into browser IndexedDB (`PrismDB`).
- **12.3.2** Trigger Dexie `useLiveQuery()` observers to update UI components reactively upon local database commits.

# Project Requirements

Hardware and software needed to build and run PrismSpace

# 16. Hardware Requirements

- **CPU:** Multi-core processor for the Next.js client, FastAPI service, and local development tools.
- **GPU:** NVIDIA RTX 3050 or above for embedding generation, vector indexing, and model-training workloads.
- **Memory:** Minimum 16 GB RAM for running the frontend, backend, databases, and supporting services concurrently.
- **Storage:** SSD with sufficient capacity for the codebase, PrismDB data, logs, model checkpoints, and FAISS indexes.
- **Network:** Reliable internet connection for provider APIs, MCP integrations, and model downloads.
- **Security:** Secure environment-variable or secret-manager support for provider and tool credentials.

# 17. Software Requirements

- **Next.js 14 / React / TypeScript:** browser-based workspace for objectives, chat history, agent status, and approvals.
- **FastAPI / Python:** asynchronous backend framework for agent orchestration, REST endpoints, and runtime events.
- **Dexie.js / IndexedDB:** local ACID storage for sessions, messages, files, settings, and tool history.
- **Joblib / scikit-learn:** Python libraries for data preparation, utility functions, serialization, and general machine-learning support.
- **MCP JSON-RPC:** MCP (Model Context Protocol) uses JSON-RPC (JavaScript Object Notation–Remote Procedure Call) messages to standardize tool names, inputs, outputs, and errors for browser, terminal, filesystem, GitHub, database, and workspace tools.
- **Provider SDKs/APIs: NVIDIA NIM** provides optimized model-serving endpoints; the **Groq SDK** supports low-latency inference; and the **Anthropic Claude SDK/API**, OpenAI-compatible APIs, and similar provider clients connect the orchestrator to different language models.

# Project Implementation

Frontend, backend, and training components

# 18. Frontend Implementation

- **Next.js 14:** React-based framework for building the browser application and managing frontend pages.
- **React:** component library for creating reusable user-interface elements.
- **TypeScript:** typed JavaScript used for safer and more maintainable frontend development.
- **Tailwind CSS:** utility-first styling framework for responsive layouts and consistent visual design.
- **Dexie.js, IndexedDB, and dexie-react-hooks:** browser storage and reactive data-access libraries for local application data.
- **Lucide React:** icon library used for interface symbols and controls.
- **Framer Motion and GSAP (GreenSock Animation Platform):** frontend animation libraries for smooth interface transitions.

# 19. Backend Implementation

- **FastAPI:** Python framework for building asynchronous backend services and application endpoints.
- **Python:** primary backend language for API services, integrations, data handling, and runtime coordination.
- **Uvicorn:** lightweight **ASGI (Asynchronous Server Gateway Interface)** server used to run FastAPI. ASGI connects asynchronous Python applications to web servers and supports non-blocking concurrent requests.
- **REST (Representational State Transfer):** API style for communicating with the frontend through structured HTTP requests and responses.
- **Pydantic:** schema validation and typed data models for request, response, configuration, and tool payloads.
- **MCP (Model Context Protocol) and JSON-RPC (JavaScript Object Notation–Remote Procedure Call):** standards for structured communication between the backend and external tools.
- **Provider SDKs/APIs:** integration layer for connecting the backend to external language-model and model-serving services.

# 20. Model Training Modules

- **Dataset preparation:** collects, cleans, and combines task traces and preference data using **Pandas**, **NumPy**, and **PyArrow**.
- **Intent classification:** categorizes user objectives using **scikit-learn** text features, classifiers, and **XGBoost**.
- **Agent routing:** selects specialist agents using shared **scikit-learn** pipelines and **XGBoost** classification support.
- **Model and provider routing:** compares task features and provider signals using **Pandas**, **NumPy**, **scikit-learn**, and **XGBoost**.
- **Performance prediction:** estimates latency and cost with **XGBoost** regression, supported by **scikit-learn** preprocessing.
- **Safety prediction:** estimates approval needs and detects unusual patterns with **scikit-learn**, **XGBoost**, and **Isolation Forest** utilities.
- **Retrieval:** uses **FAISS (Facebook AI Similarity Search)**, **NumPy**, and optional **sentence-transformers** embeddings for context search.
- **Preference alignment:** uses **PyTorch**, **Hugging Face Transformers**, **TRL (Transformer Reinforcement Learning)**, **PEFT**, and **Datasets** for ORPO training.
- **Runtime artifacts:** uses **Joblib**, **Safetensors**, and **Hugging Face Accelerate** to save, validate, and load trained components.

**Artifact:** a saved output from training, such as a trained model, vector index, or configuration file, that can be reused later. **FAISS (Facebook AI Similarity Search):** a library for quickly finding similar vectors. **PEFT (Parameter-Efficient Fine-Tuning):** adapts a model by training only a small set of additional parameters.

# System Design

Modules, responsibilities, and system-level organization

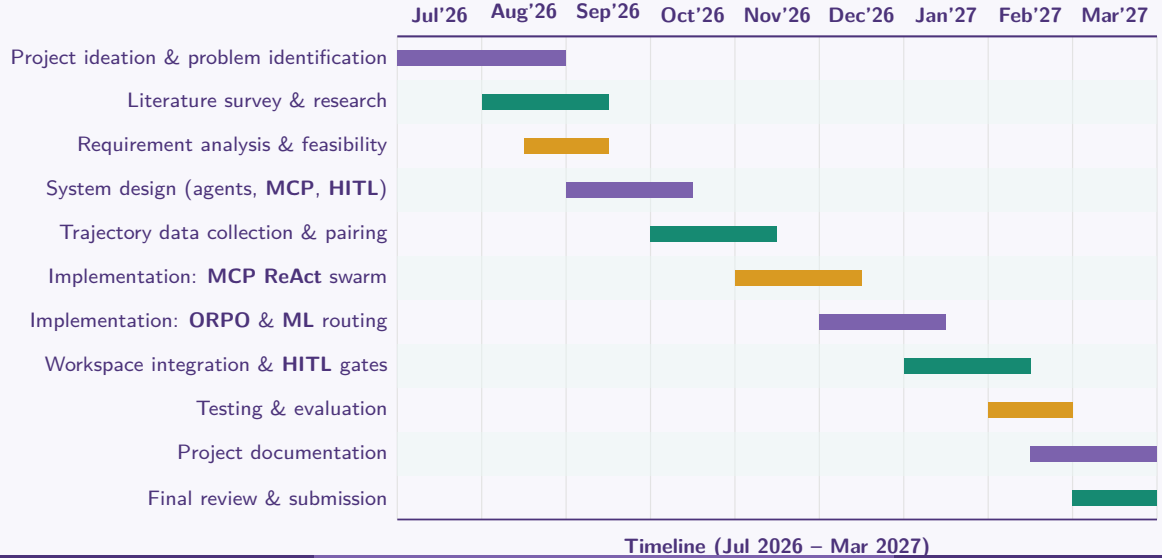
# 21. System Modules and Responsibilities

- **User and Session Module:** captures objectives, settings, conversation history, uploaded files, and session state.
- **Context and Routing Module:** classifies intent and selects the most suitable provider, model, and specialist agents.
- **DAG (Directed Acyclic Graph) Planning Module:** decomposes objectives into dependency-aware subtasks and schedules parallel work.
- **Agent Execution Module:** runs bounded **ReAct (Reasoning and Acting)** loops and invokes approved tools.
- **MCP (Model Context Protocol) Integration Module:** exposes filesystem, terminal, database, memory, and external-service tools through **JSON-RPC (JavaScript Object Notation–Remote Procedure Call)**.
- **Verification and Safety Module:** applies schema checks, **ORPO (Odds Ratio Preference Optimization)** preference gating, anomaly detection, and **HITL (Human-in-the-Loop)** approvals.
- **Persistence Module:** stores sessions, messages, artifacts, and errors in the local **PrismDB (browser IndexedDB database)**.

# 22. Conclusion

- **PrismSpace** converts a user's natural-language goal into smaller, organized tasks.
- Independent tasks can be executed by multiple agents in parallel.
- Standard tools are connected through the **Model Context Protocol (MCP)**.
- Human approval and local browser storage improve safety, privacy, and control.
- The platform provides a flexible foundation for future agent-based applications.

# 23. Gantt Chart



# References


[1] **Cornalba F, Disselkamp C, Scassola D, Helf C (2024)** Multi-objective reward generalization: improving performance of Deep Reinforcement Learning for applications in single-asset trading. *Neural Computing and Applications*. DOI: 10.1007/s00521-023-09033-7

[2] **Schneckenreither M, Moser G (2025)** Average reward adjusted discounted reinforcement learning. *Neural Computing and Applications*. DOI: 10.1007/s00521-024-10620-5

[3] **Aina K, Ha S (2026)** Deep reinforcement learning for multi-agent coordination. *Artificial Life and Robotics*. DOI: 10.1007/s10015-025-01089-z

[4] **Rother D, Pajarinen J, Peters J, Weisswange TH (2025)** Open-ended coordination for multi-agent systems using modular open policies. *Autonomous Agents and Multi-Agent Systems*. DOI: 10.1007/s10458-025-09723-7

[5] **Liu D, Ren F, Yan J, Su G, Gu W, Kato S (2026)** Improving scalability of multi-agent deep reinforcement learning with suboptimal human knowledge. *Autonomous Agents and Multi-Agent Systems*. DOI: 10.1007/s10458-025-09729-1

[6] **Marzari L, Liu C, Donti PL, Marchesini E (2026)** $\varepsilon$-retraining reinforcement learning algorithms. *Autonomous Agents and Multi-Agent Systems*. DOI: 10.1007/s10458-026-09748-6

[7] **Wong A, Bäck T, Kononova AV, Plaat A (2023)** Deep multiagent reinforcement learning: challenges and directions. *Artificial Intelligence Review*. DOI: 10.1007/s10462-022-10299-x

[8] **Niu B, Luo B, Cui Y, Xu X, Zhao Y, Feng Y (2025)** HIAT: human-in-the-loop reinforcement learning with auxiliary task. *Artificial Intelligence Review*. DOI: 10.1007/s10462-025-11260-4

# References


[9] **Lee J (2025)** InstructPatentGPT: training patent language models to follow instructions with human feedback. *Artificial Intelligence and Law*. DOI: 10.1007/s10506-024-09401-1

[10] **Zhuang S, Qiao Z, Zuccon G (2023)** Reinforcement online learning to rank with unbiased reward shaping. *Information Retrieval Journal*. DOI: 10.1007/s10791-022-09413-y

[11] **Ghazal TM, Awouda AE, Hasan MK, Rahman AHA, Islam S, Saeed RA, Elshafie H, Iqbal A, Hussein AH (2025)** Knowledge Learning for Securing Workflow Scheduling Algorithm in Mobile Edge Computing. *Mobile Networks and Applications*. DOI: 10.1007/s11036-025-02465-6

[12] **Barredo P, Puente J (2025)** Energy-aware cooperative multi-fitness evolutionary algorithm for workflow scheduling in cloud computing. *Natural Computing*. DOI: 10.1007/s11047-025-10023-y

[13] **Heirati A, Fartash M, Khalily-Dermany M (2025)** Optimized Task Scheduling in Fog-Cloud Computing Using Hybrid Deep Learning and Metaheuristic Algorithms. *Neural Processing Letters*. DOI: 10.1007/s11063-025-11819-w

[14] **Abdi S, Ashjaei M, Mubeen S (2024)** Cost-aware workflow offloading in edge-cloud computing using a genetic algorithm. *The Journal of Supercomputing*. DOI: 10.1007/s11227-024-06341-0

[15] **Xia Y, Zhu J, Zhu L (2023)** Dynamic role discovery and assignment in multi-agent task decomposition. *Complex & Intelligent Systems*. DOI: 10.1007/s40747-023-01071-x

[16] **Li J, Xing L, Zhong W, Cai Z, Hou F (2023)** Decision variable contribution based adaptive mechanism for evolutionary multi-objective cloud workflow scheduling. *Complex & Intelligent Systems*. DOI: 10.1007/s40747-023-01137-w

# References


[17] **Yu X, Lin Y, Wang X, Han S, Lv K (2024)** GHQ: grouped hybrid Q-learning for cooperative heterogeneous multi-agent reinforcement learning. *Complex & Intelligent Systems*. DOI: 10.1007/s40747-024-01415-1

[18] **Xu C, Wang J, Zhu X, Yue Y, Zhou W, Liang Z, Wojtczak D (2024)** Decentralized multi-agent cooperation via adaptive partner modeling. *Complex & Intelligent Systems*. DOI: 10.1007/s40747-024-01421-3

[19] **Vyhmeister E, Castañé GG, Östberg PO, Thevenin S (2023)** A responsible AI framework: pipeline contextualisation. *AI and Ethics*. DOI: 10.1007/s43681-022-00154-8

[20] **Cavalcante Siebert L, Lupetti ML, Aizenberg E, Beckers N, Zgonnikov A, Veluwenkamp H, Abbink D, Giaccardi E, Houben GJ, Jonker CM, van den Hoven J, Forster D, Lagendijk RL (2023)** Meaningful human control: actionable properties for AI system development. *AI and Ethics*. DOI: 10.1007/s43681-022-00167-3

[21] **Yue Q, Tian L, Feng X, Yuan Z, Wei S, Xia J, Chen B, Song X, Yao Y (2026)** CPN-HRL: a hierarchical deep reinforcement learning approach for priority-aware task scheduling in CPN enabled by cloud-edge-end environments. *Journal of King Saud University Computer and Information Sciences*. DOI: 10.1007/s44443-026-00690-x

[22] **Skirzyński J, Jain Y, Lieder F (2023)** Automatic discovery and description of human planning strategies. *Behavior Research Methods*. DOI: 10.3758/s13428-023-02062-z

[23] **Zhang S, Zhao Z, Liu C, Qin S (2023)** Data-intensive workflow scheduling strategy based on deep reinforcement learning in multi-clouds. *Journal of Cloud Computing*. DOI: 10.1186/s13677-023-00504-9

[24] **Wang Z, Chen S, Bai L, Gao J, Tao J, Bond RR, Mulvenna MD (2023)** Reinforcement learning based task scheduling for environmentally sustainable federated cloud computing. *Journal of Cloud Computing*. DOI: 10.1186/s13677-023-00553-0

[25] **Subashree S, Rajakumaran M, Pushpa G, Senthilkumar S (2025)** Hybrid dolphin swarm sparrow search optimization based multi-objective cloud workflow scheduling. *Journal of Cloud Computing*. DOI: 10.1186/s13677-025-00828-8

# Thank You!

Nobin Sijo | Pranav P | Prasanth P | Sankar Narayan S

Department of Computer Science & Engineering
College of Engineering, Karunagappally